

June 22, 2026

Prof. Raphaël Frank
UBIX Research Group
Interdisciplinary Centre for Security, Reliability and Trust (SnT)
University of Luxembourg

Dear Prof. Raphaël Frank,

I am writing to express my strong interest in the Doctoral Researcher position focusing on Autonomous Driving Systems and human-centered interaction strategies within the AuTHUMate project. As an AI & Data Science Engineer specializing in predictive modeling for complex, high-stakes physical systems, I am highly motivated to transition my expertise into the realm of Vision Language Models (VLMs) and automated driving.

Throughout my engineering theses, my primary focus has been ensuring that neural network architectures respect the physical and temporal realities of the systems they monitor. For my Master's thesis ("Cardiovascular Medical Digital Twin"), I developed MedicalMambaTS, a continuous-time state-space model designed to process highly irregular, multimodal sensor data in real-time. I also engineered PI-DeepONet, mapping strict differential equations directly into the PyTorch Autograd graph to prevent neural hallucinations.

Similarly, in my State Engineer thesis ("Gas Turbine Digital Twin"), I designed Spatial-Temporal Transformers to assess the degradation and "readiness" of complex mechanical systems based on multidimensional SCADA telemetry. I believe these methodologies—specifically continuous-time modeling and multimodal data fusion—are highly transferable to the AuTHUMate project. Assessing a human driver's readiness, trust, and context using multimodal cabin data requires the exact same fusion of temporal sequences and computer vision that I have utilized in my previous digital twin research.

I possess strong programming skills in Python (PyTorch), a deep understanding of Control Systems and Machine Learning, and a proven track record of open-sourcing complex architectures.

Please find attached my CV, transcripts, and academic references. I would welcome the opportunity to discuss my qualifications with the UBIX team in an interview.

Sincerely,
Ismail Aissaoui
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